

ORGANISATION – THE HEART & CIRCULATORY SYSTEM

KEY VOCABULARY

Atria: the upper chambers that receive blood into the heart.

Ventricles: the lower chambers that pump blood out of the heart.

Artery: carries blood away from the heart at high pressure.

Vein: carries blood back to the heart; has valves.

COMMON MISCONCEPTIONS

~~Arteries always carry oxygenated blood.~~

✓ The pulmonary artery carries deoxygenated blood.

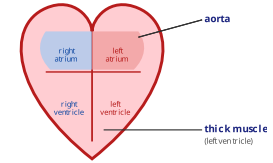
~~The heart is one single pump.~~

✓ It is a double pump — right and left sides.

~~Blood flows in both directions in veins.~~

✓ Valves keep blood flowing one way only.

STRUCTURE OF THE HEART



The **left ventricle** has the thickest muscle to pump blood around the body.

1 NAME THE CHAMBERS

Name the **four chambers** of the heart. [4 marks]

2 DOUBLE CIRCULATION

Explain what is meant by a 'double circulatory system'. [2 marks]

3 THICK MUSCLE

Explain why the left ventricle wall is thicker than the right. [2 marks]

4 BLOOD VESSELS

Complete the sentence using the word bank. [2 marks]

_____ carry blood away from the heart and have _____ muscular walls.

arteries

veins

thick

valves

5 HEART RATE

The heart beats 72 times per minute. **Calculate** how many beats this is in one hour. [2 marks]

Answer: _____ beats

TIP: Right side = lungs (pulmonary); left side = rest of body (systemic).

6 THE PACEMAKER

Name the group of cells that control the natural resting heart rate. [1 mark]

✓ Think: Why must blood pass through the heart twice in one full circuit?